PEAKS OVER THRESHOLD IN PRACTICE

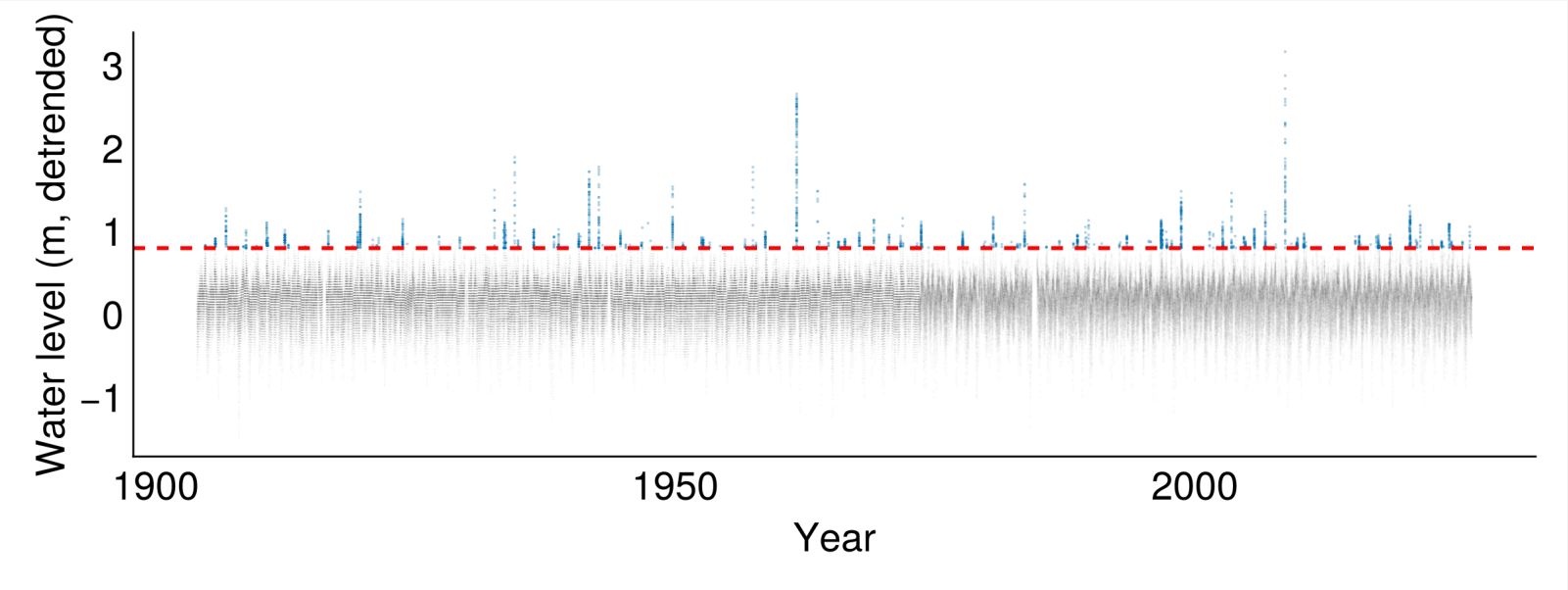
James Doss-Gollin

CEVE 543

2026-09-21



HOURLY WATER LEVELS AT GALVESTON, TX



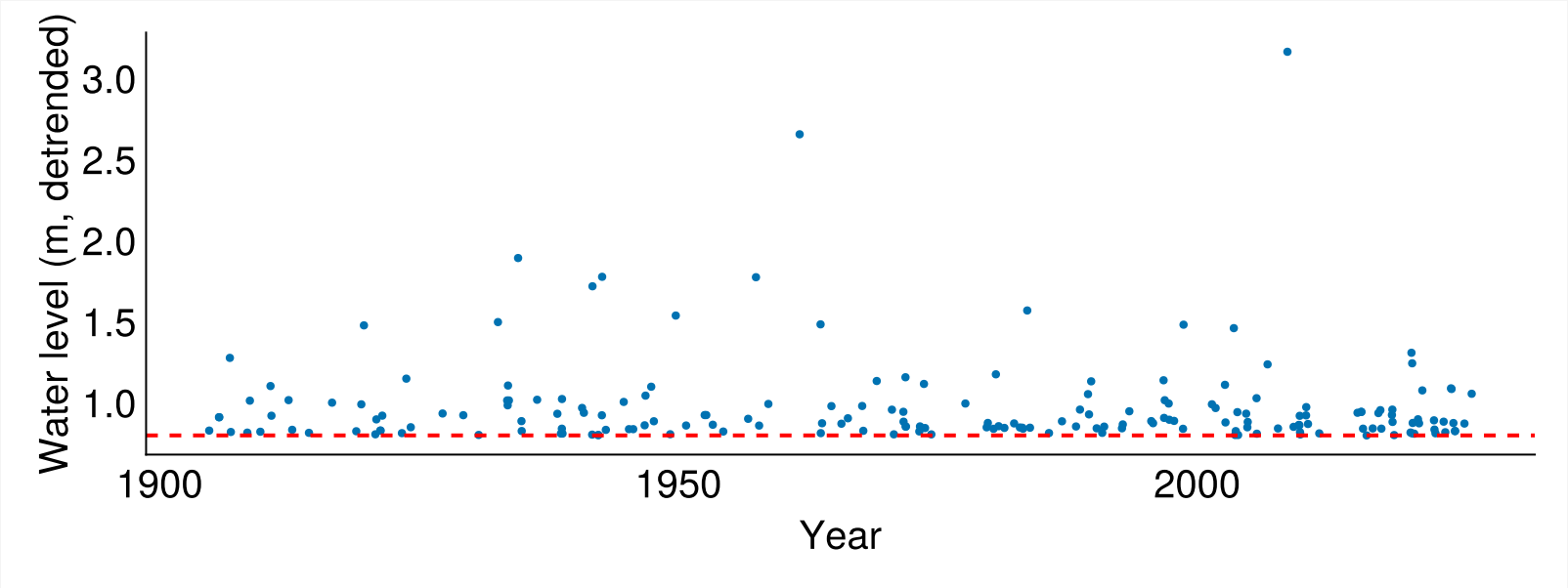
- Galveston Pier 21
- 120 years of hourly readings (detrended by annual mean sea level).

ONE WAY TO DECLUSTER: THE GAP RULE

1. Group consecutive exceedances into **clusters**: a new cluster starts when the gauge has been below the threshold for at least r hours.
2. Keep only the **maximum** of each cluster.

Here $r = 72$ (three days).

AFTER DECLUSTERING



The GPD parameters and the Poisson rate move in opposite directions when the threshold or gap parameter changes, so the return level is approximately insensitive to these choices.

CHOOSING A THRESHOLD

We picked 0.8 m, but why not 0.5 or 1.2?

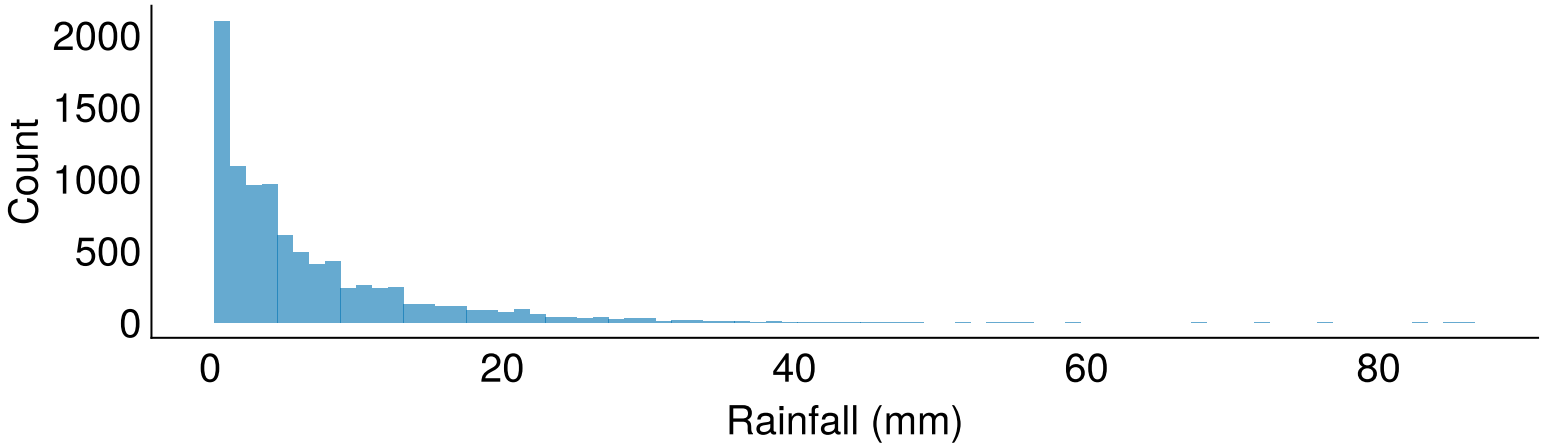
- **Too low:** the GPD is an asymptotic approximation, and it gets worse far from the tail. More data, but more bias.
- **Too high:** fewer exceedances, so the parameter estimates are noisier. Less bias, but more variance.

We want the **lowest threshold where the GPD holds**. Two diagnostic tools help us check.

We will look at each on the textbook example first, then on Galveston.

THE COLES RAINFALL DATA

17,531 daily rainfall totals from a single station (Coles, 2001, §4.4.1). Also documented in `Extremes.jl` tutorial.



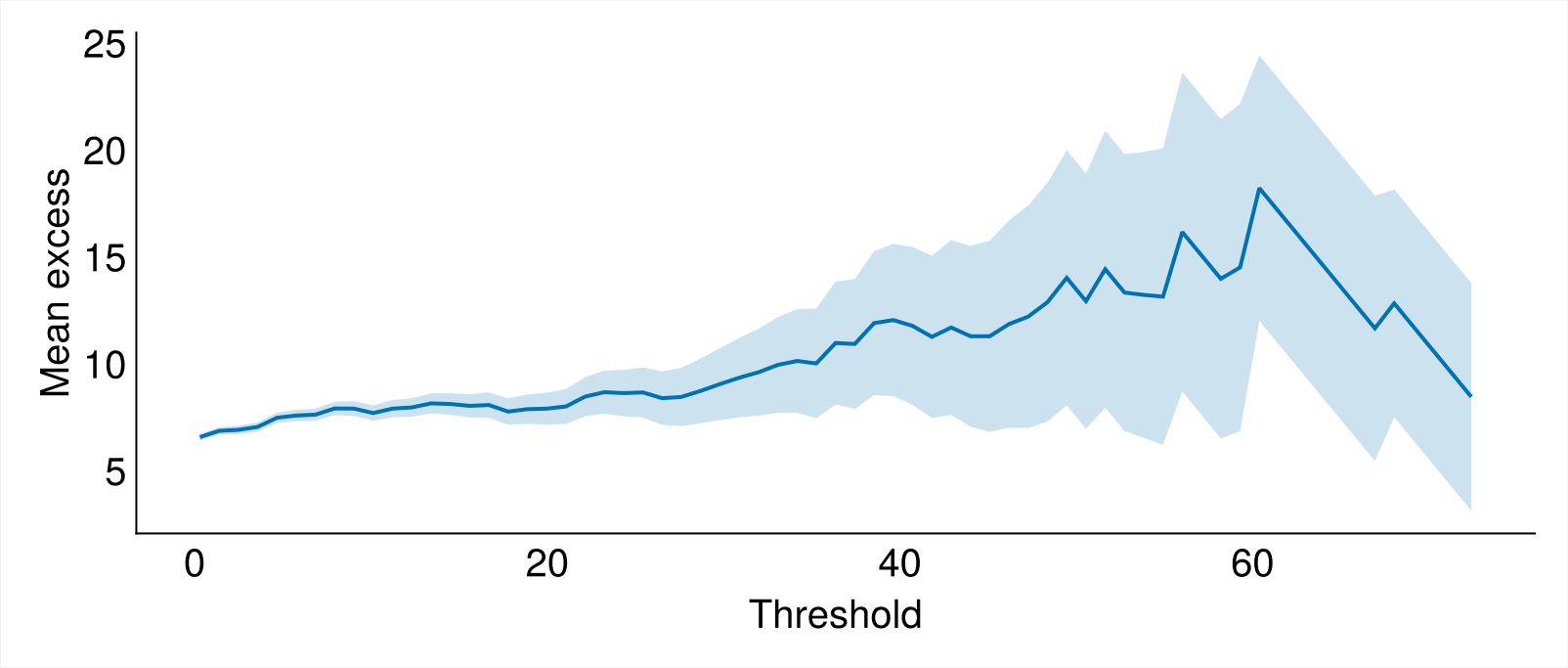
MEAN RESIDUAL LIFE PLOT

At each candidate threshold u, compute the average excess

$$\hat{E}[X - u \mid X > u]$$

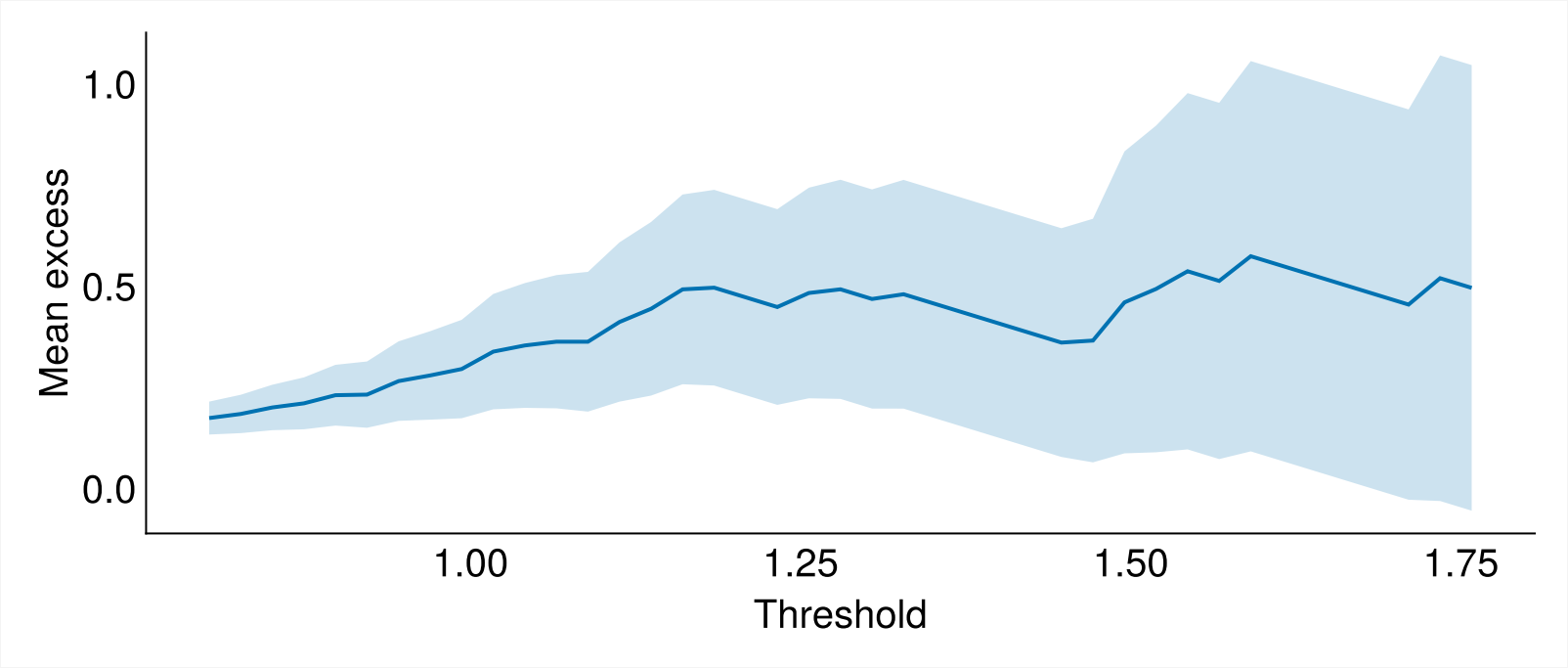
and plot it against u. If the GPD holds above u_0, the mean excess is linear in u for $u > u_0$.

MEAN RESIDUAL LIFE PLOT



Roughly linear from about 30 mm onward, until it gets “too large” to have much data.

MEAN RESIDUAL LIFE: GALVESTON



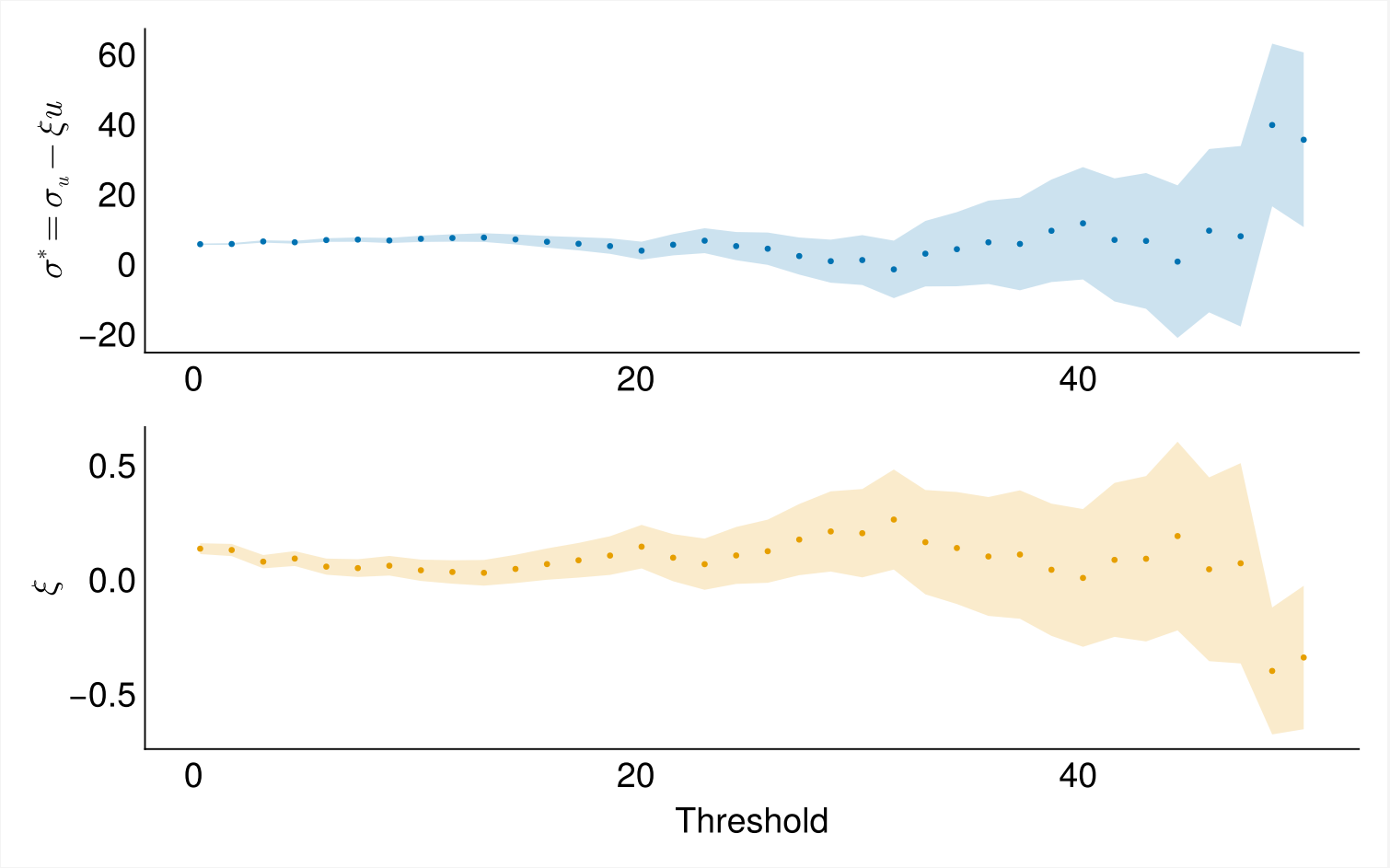
Less clear. Coles: MRL plots “can be difficult to interpret in practice” (p. 79).

PARAMETER STABILITY: WHAT WE ARE PLOTTING

Fit the GPD at each candidate threshold:

- If the GPD holds above u_0, the shape ξ should be constant for all $u > u_0$.
- The scale σ_u changes with u, so we plot the reparameterized $\sigma^* = \sigma_u - \xi u$, which should also be constant.

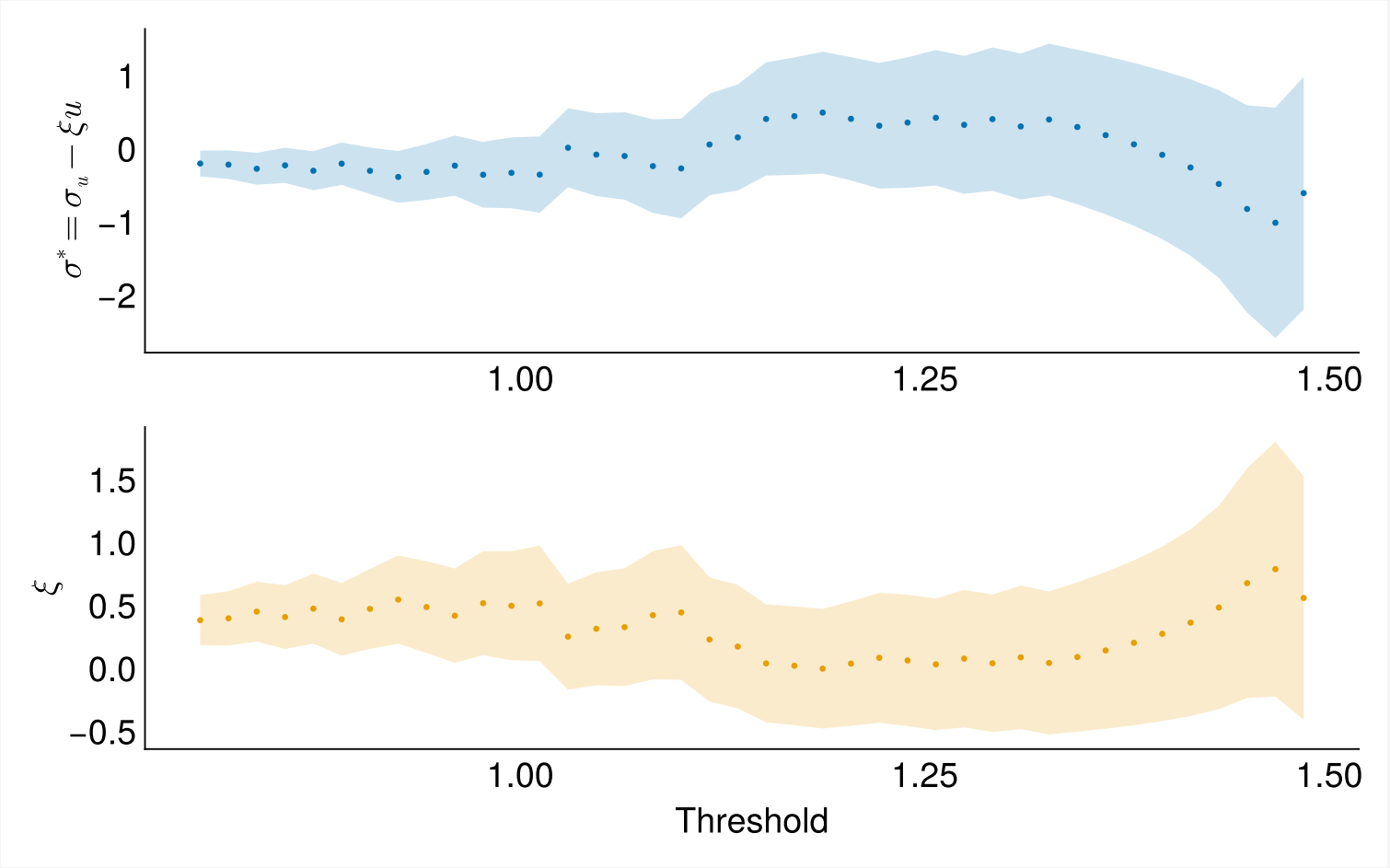
PARAMETER STABILITY: RAINFALL



PARAMETER STABILITY: RAINFALL

Both σ^* and ξ are stable from about 30 mm onward. Coles fits at $u = 30$ and gets $\hat{\sigma} = 7.44$, $\hat{\xi} = 0.184$ with 152 exceedances.

PARAMETER STABILITY: GALVESTON



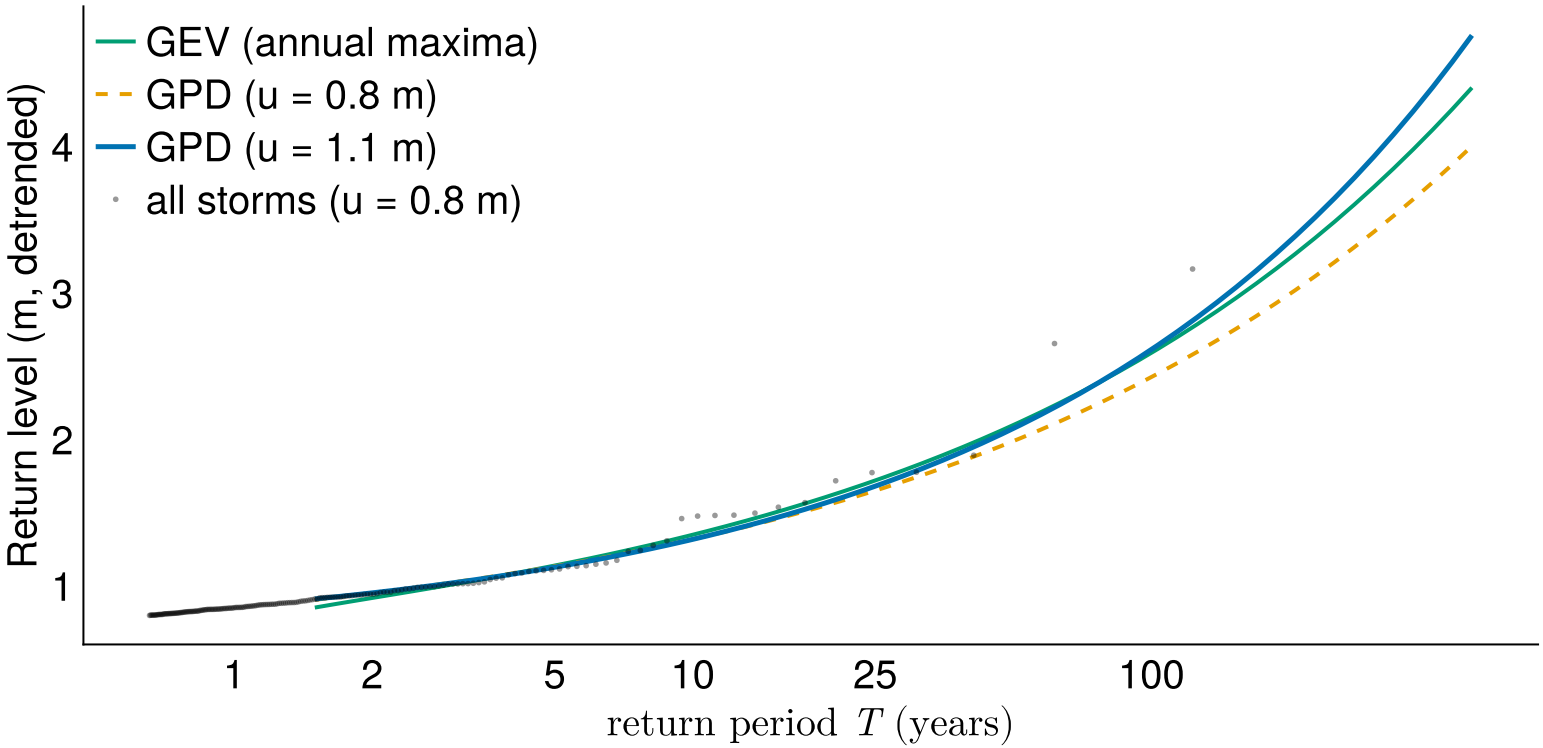
PARAMETER STABILITY: GALVESTON

ξ looks stable from about 1.1 m onward. Messier than the rainfall case, but a defensible choice.

FITTING AND COMPARING: GALVESTON

```
Threshold: 1.1 m
Cluster maxima: 28 over 123 years (λ = 0.23/yr)
GPD fit: σ = 0.220, ξ = 0.452
```

RETURN LEVELS: THREE ESTIMATES



RETURN LEVELS: THREE ESTIMATES

Three estimates of the same return level curve, from the same detrended record. The empirical points are every declustered storm above 0.8 m, plotted at $T = N_{\text{years}}/\text{rank}$. The two GPD fits and the GEV agree at moderate return periods and diverge in the tail.